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Shehu Integral Transform and Hyers-Ulam Stability of n^{th} order Linear Differential Equations

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ABSTRACT

In this paper, we establish the Shehu transform expression for homogeneous and non-homogeneous linear differential equations. With the help of this new integral transform, we solve higher order linear differential equations in the Shehu sense. Moreover, this paper introduced a new concept to find the Hyers-Ulam stability of the differential equation. This is the first attempt to use the Shehu transform to prove the Hyers-Ulam stability of the differential equation. Finally, the applications and remarks are discussed to demonstrate our strategy. Applications of the Shehu transform to fractional differential equations, Newton's law of cooling, and free undamped motion are also discussed.

Introduction

In 1823, Niels Henrik Abel used Abel's integral equation in mathematical form as [1–3] to discuss the motion of particles on a smooth curve sitting on a vertical plane

$$f(x) = \int_0^x \frac{1}{\sqrt{x-\check{e}}} g(\check{e}) d\check{e}, \tag{1.1}$$

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where the kernal of (1.1), $k(x,\check{e}) = \frac{1}{\sqrt{x-\check{e}}}$ becomes ∞ at $\check{e} = x$, the function $g(\check{e})$ is unknown function. The homotopy analysis transform method [4], Taylor expansion method [5,7], homotopy regularization method [6] and matrix method by Genocchi polynomials [8] have been applied for solving problem (1.1).

Integral transforms are widely used mathematical techniques in modern times for solving advanced problems in engineering and science, and they are mathematically demonstrated in terms of differential equations with constant or variable coefficients, partial differential equations with constant or variable coefficients, integral equations, partial integro-differential equations, integro-differential equations, and so on. A variety of experts have used integral transforms (Laplace transform, Fourier transform, Kamal transform, Hankel transform, Mohand transform, Elzaki transform, Sumudu transform, Aboodh transform, Wavelet transform, and so on) to solve a variety of problems in science, engineering, and everyday life (see [9–14]). Veeresha recently investigated the solution of fractional differential equations in a variety of fields, including the Laguerre polynomial - based operational matrix of integration, the computational framework for fractional Gross–Pitaevskii equations, the dynamics of a fractional epidemiological model with disease infection, and so on [15–21].

Currently, researchers are focusing their efforts on creating numerical and analytical methodologies for tackling fractional-order issues [22–25]. As a result, a number of analytical and numerical approaches to solving fractional models have been developed and put into use. The history of integral transforms may be traced back to the 1780s, when Laplace started working on one, and 1822, when Joseph Fourier started working on one. Integrals are one of the most valuable and effective methodologies in theoretical and applied mathematics, with applications in quantum physics, mechanical engineering, and a wide range of other subjects. In addition, many models in chemistry, architecture, and other social sciences employ the integral transform to analyse them [26].

In recent years, different integral transforms, such as Laplace transform [27–29], Fourier transform [30,31], Hankel transform [32], Mellin transform [33], Z-transform [34], Elzaki transform [35], Sumudu transform [36], Hermite transform [37] etc. have been used for the solution of different physical models.

The Shehu transform of the function $Ƒ(\check{e}),\ \forall\ \check{e} \geq 0$ of exponential order is characterized as

$$S\{Ƒ(\check{e})\} = \int_0^\infty Ƒ(\check{e}) e^{-\frac{v\check{e}}{u}} d\check{e} = Ḩ(v,u), v > 0, u > 0, \tag{1.2}$$

where the operator S is called the Shehu transform operator.

If $Ƒ(\check{e})$ is piecewise continuous, the function's Shehu transform exists for $\check{e} \geq 0$. These are the only requirements for the existence of Shehu transforms in the function $Ƒ(\check{e})$. The meaning of the Shehu transform, a few relevant properties of the Shehu transform, the Shehu transform of function derivatives, the inverse Shehu transform, and the linearity property of the inverse Shehu transform are all discussed in this work.

In quantum calculus, Sinha and Panda [38] the Shehu transform. Alfaqeih [39] uses double Shehu transforms to solve Caputo fractional partial differential equations. Al-Qurashi [40] reported new Shehu decomposition method computations for the two-mode version of the fractional Zakharov-Kuznetsov model in plasma fluid. The Shehu transform approach for solving fuzzy differential equations with fractional and integer order derivatives is investigated by [41,42].

Obloza studied the stability of linear differential equations and the connections between Hyers and Lyapunov stability in ordinary differential equations ([43,44]). Takahasi [45] and Alsina [46] looked into the Hyers-Ulam stability of the Banach space-valued differential equation $y' = \lambda y$, as well as various inequalities and stability results for the exponential function. For examples of current uses of Hyres-Ulam stability results, see [47–51].

As per the above results, the authors prove the various type of Hyres-Ulam stability of the following n^{th} order linear differential equations

$$Ƒ^n(\check{e}) + a_1(\check{e})Ƒ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ƒ'(\check{e}) + a_n(\check{e})Ƒ(\check{e}) = 0 \tag{1.3}$$

and

$$Ƒ^n(\check{e}) + a_1(\check{e})Ƒ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ƒ'(\check{e}) + a_n(\check{e})Ƒ(\check{e}) = m(\check{e}) \tag{1.4}$$

by using Shehu transform method.

This paper's outline includes the following: The properties of the Shehu transform are explored in Section 2, including linearity, change of scale, shifting, Shehu transform of derivatives, convolution, and Shehu transform of frequently encountered functions. In Section 3, the inverse Shehu transform and the essential definitions of Hyers-Ulam stability are studied. The authors investigated various types of stability, such as Hyers-Ulam stability, Hyers-Ulam σ-stability, Mittag-Leffler-Hyers-Ulam stability, and Mittag-Leffler-Hyers-Ulam σ-stability of n^{th} order linear differential equations, using the Shehu transform method in Sections 4 and 5. The Shehu transform method's applications are explained in Section 6.

Some useful properties of Shehu Transform

In this section, the authors discuss some basic properties of the Shehu transform, such as linearity, Shehu transform of derivatives, the convolution theorem and so on.

Linearity

Let $Ḩ_1(\nu,u)$ and $Ḩ_2(\nu,u)$ are Shehu transforms of the functions $Ƒ_1(\check{e})$ and $Ƒ_2(\check{e})$ respectively. Then $[aƑ_1(\check{e})+bƑ_2(\check{e})] = [aḨ_1(\nu,u)+bḨ_2(\nu,u)]$, where a,b are arbitrary constants.

Proof. The Shehu transform of the function $Ƒ(\check{e})$ is defined by

$$S\{Ƒ(\check{e})\} = \int_0^\infty Ƒ(\check{e})e^{-\frac{\nu\check{e}}{u}}d\check{e} = Ḩ(\nu,u).$$

Then

$$\begin{aligned} S\{aƑ_1(\check{e})+bƑ_2(\check{e})\} &= \int_0^\infty [aƑ_1(\check{e})+bƑ_2(\check{e})]e^{-\frac{\nu\check{e}}{u}}d\check{e} \\ &= a\int_0^\infty Ƒ_1(\check{e})e^{-\frac{\nu\check{e}}{u}}d\check{e} + b\int_0^\infty Ƒ_2(\check{e})e^{-\frac{\nu\check{e}}{u}}d\check{e} \\ S\{aƑ_1(\check{e})+bƑ_2(\check{e})\} &= aḨ_1(\nu,u)+bḨ_2(\nu,u), \end{aligned}$$

where a,b are arbitrary constants. □

Change of scale

If $Ḩ(\nu,u)$ be the Shehu transform of the function $Ƒ(\check{e})$, then $Ƒ(a\check{e}) = \frac{1}{a}Ḩ\left(\frac{\nu}{a},u\right)$.

Proof. The Shehu transform of the function $Ƒ(\check{e})$ is defined by

$$S\{Ƒ(\check{e})\} = \int_0^\infty Ƒ(\check{e})e^{-\frac{\nu\check{e}}{u}}d\check{e} = Ḩ(\nu,u).$$

Put $\check{e} = \frac{p}{a} \Rightarrow d\check{e} = \frac{dp}{a}$, we have

$$\begin{aligned} S\{Ƒ(\check{e})\} &= \int_0^\infty Ƒ(\check{e})e^{-\frac{\nu}{u}\left(\frac{p}{a}\right)}\frac{dp}{a} \\ &= \frac{1}{a}\int_0^\infty Ƒ(\check{e})e^{-\left(\frac{\nu}{ua}\right)p}dp \\ S\{Ƒ(\check{e})\} &= \frac{1}{a}H\left(\frac{\nu}{a},u\right). \end{aligned}$$

□

Shifting property

Let $Ḩ(\nu,u)$ be the Shehu transform of the function $Ƒ(\check{e})$. Then $e^{a\check{e}}Ƒ(\check{e}) = Ḩ(\nu-au,u)$.

Proof. The Shehu transforms of the function $Ƒ(\check{e})$ is defined by

$$S\{Ƒ(\check{e})\} = \int_0^\infty Ƒ(\check{e})e^{-\frac{\nu\check{e}}{u}}d\check{e}.$$

Then

$$\begin{aligned} S\{e^{a\check{e}}Ƒ(\check{e})\} &= \int_0^\infty Ƒ(\check{e})e^{a\check{e}}e^{-\frac{\nu\check{e}}{u}}d\check{e} \\ &= \int_0^\infty Ƒ(\check{e})e^{-\left(\frac{\nu-au}{u}\right)\check{e}}d\check{e} \\ &= Ḩ(\nu-au,u). \end{aligned}$$

□

Shehu transform of derivatives

If $S\{Ƒ(\check{e})\} = Ḩ(\nu,u)$, then

(a) $S\{Ƒ'(\check{e})\} = \frac{\nu}{u}Ḩ(\nu,u) - Ƒ(0)$
(b) $S\{Ƒ''(\check{e})\} = \frac{\nu^2}{u^2}Ḩ(\nu,u) - \frac{\nu}{u}Ƒ(0) - Ƒ'(0)$
(c) $S\{Ƒ^n(\check{e})\} = \frac{\nu^n}{u^n}Ḩ(\nu,u) - \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ƒ^k(0).$

Proof:

1. The Shehu transform of the Eq. (1.2) gives

$$S\{F̧(\breve{e})\} = \int_0^\infty F̧(\breve{e}) e^{-\frac{\nu\breve{e}}{u}} d\breve{e}$$
$$S\{F̧'(\breve{e})\} = \int_0^\infty F̧'(\breve{e}) e^{-\frac{\nu\breve{e}}{u}} d\breve{e}.$$

By using rule of integration, we get

$$S\{F̧'(\breve{e})\} = -F̧(0) + \frac{\nu}{u} Ḩ(\nu, u)$$
$$S\{F̧'(\breve{e})\} = \frac{\nu}{u} Ḩ(\nu, u) - F̧(0).$$

2. Since $S\{F̧'(\breve{e})\} = \frac{\nu}{u} Ḩ(\nu, u) - F̧(0)$. Then

$$S\{F̧''(\breve{e})\} = \int_0^\infty F̧''(\breve{e}) e^{-\frac{\nu\breve{e}}{u}} d\breve{e}.$$

By using rule of integration, we get

$$\begin{aligned} S\{F̧''(\breve{e})\} &= -F̧'(0) + \frac{\nu}{u} S\{F̧'(\breve{e})\} \\ &= -F̧'(0) + \frac{\nu}{u}\left(\frac{\nu}{u} Ḩ(\nu, u) - F̧(0)\right) \\ S\{F̧''(\breve{e})\} &= \frac{\nu^2}{u^2} Ḩ(\nu, u) - \frac{\nu}{u} F̧(0) - F̧'(0). \end{aligned}$$

3. In general, for any positive integer n, we have

$$S\{F̧^n(\breve{e})\} = \frac{\nu^n}{u^n} Ḩ(\nu, u) - \sum_{k=0}^{n-1} \left(\frac{\nu}{u}\right)^{n-(k+1)} F̧^k(0).$$

Convolution theorem for Shehu transform

If Shehu transform of the functions $F̧_1(\breve{e})$ and $F̧_2(\breve{e})$ are $Ḩ_1(\nu, u)$ and $Ḩ_2(\nu, u)$ respectively, then the Shehu transform of their convolution $F̧_1(\breve{e}) \star F̧_2(\breve{e})$ is given by

$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = S\{F̧_1(\breve{e})\} S\{F̧_2(\breve{e})\} = Ḩ_1(\nu, u) Ḩ_2(\nu, u),$$

where $F̧_1(\breve{e}) \star F̧_2(\breve{e}) = \int_0^{\breve{e}} F̧_1(\breve{e} - x) F̧_2(x) dx = \int_0^{\breve{e}} F̧_1(x) F̧_2(\breve{e} - x) dx.$

Proof. By the definition of Shehu transform, we have

$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = \int_0^\infty e^{-\frac{\nu\breve{e}}{u}} [F̧_1(\breve{e}) \star F̧_2(\breve{e})] d\breve{e}$$
$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = \int_0^\infty e^{-\frac{\nu\breve{e}}{u}} \left[\int_0^\infty F̧_1(\breve{e} - x) F̧_2(\breve{e}) dx\right] d\breve{e}.$$

By changing the order of integration, we get

$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = \int_0^\infty F̧_2(\breve{e}) \left[\int_0^\infty e^{-\frac{\nu\breve{e}}{u}} F̧_1(\breve{e} - x) d\breve{e}\right] dx.$$

Put $\breve{e} - x = p$ so that $d\breve{e} = dp$ in above equation, we obtain

$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = \int_0^\infty F̧_2(\breve{e}) \left[\int_0^\infty e^{-\frac{\nu(p-x)}{u}} F̧_1(p) dp\right] dx$$
$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = \int_0^\infty F̧_2(\breve{e}) e^{-\frac{x\nu}{u}} \left[\int_0^\infty e^{-\frac{p\nu}{u}} F̧_1(p) dp\right] dx$$
$$S\{F̧_1(\breve{e}) \star F̧_2(\breve{e})\} = Ḩ_1(\nu, u) Ḩ_2(\nu, u).$$

□

Shehu transform of simple functions

In this section, we find inverse Shehu transform of simple functions.

(i) Let $\digamma(\check{e}) = 1$. then by definition of Shehu transform, we have

$$S\{\digamma(\check{e})\} = \int_0^\infty \digamma(\check{e}) e^{-\frac{v\check{e}}{u}} d\check{e}$$
$$S\{1\} = \int_0^\infty 1 e^{-\frac{v\check{e}}{u}} d\check{e}$$
$$S\{1\} = \frac{u}{v}.$$

(ii) If $\digamma(\check{e}) = \check{e}$, then by definition of Shehu transform, we get

$$S\{\check{e}\} = \int_0^\infty \check{e} e^{-\frac{v\check{e}}{u}} d\check{e}.$$

Using the definition of order of integration, we obtain

$$S\{\check{e}\} = \frac{u}{v}\int_0^\infty e^{-\frac{v\check{e}}{u}} d\check{e}$$
$$S\{\check{e}\} = \left(\frac{u}{v}\right)\left(\frac{u}{v}\right)$$
$$S\{\check{e}\} = \left(\frac{u}{v}\right)^2.$$

(iii) Let $\digamma(\check{e}) = e^{a\check{e}}$. From the definition of Shehu transform, we get

$$S\{e^{a\check{e}}\} = \int_0^\infty e^{a\check{e}} e^{-\frac{v\check{e}}{u}} d\check{e}.$$

Using the definition of order of integration, we have

$$S\{e^{a\check{e}}\} = -\frac{au^2}{a^2u^2 - vau}$$
$$= \frac{-au^2}{au(au - v)}$$
$$S\{e^{a\check{e}}\} = -\frac{u}{v - au}.$$

Inverse Shehu transform

If $\c{H}(v,u)$ is the Shehu transforms of the function $\digamma(\check{e})$, then $\digamma(\check{e})$ is called the inverse Shehu transform of $\c{H}(v,u)$ and it can be expressed mathematically as $\digamma(\check{e}) = S^{-1}\{\c{H}(v,u)\}$, where S^{-1} is an operator known as the inverse Shehu transform operator.

Linearity

If $S^{-1}\{\c{H}_1(v,u)\} = \digamma(\check{e})$ and $S^{-1}\{\c{H}_2(v,u)\} = \c{G}(\check{e})$, then

$$S^{-1}\{a\c{H}_1(v,u) + b\c{H}_2(v,u)\} = aS^{-1}\{\c{H}_1(v,u)\} + bS^{-1}\{\c{H}_2(v,u)\}$$
$$S^{-1}\{a\c{H}_1(v,u) + b\c{H}_2(v,u)\} = a\digamma(\check{e}) + b\c{G}(\check{e}),$$

where a, b are arbitrary constants.

The inverse Shehu transform of some simple functions is given in Table 2.

Definition 3.1. The Mittag Leffler function of one parameter is defined by $E_\beta(\check{e}) = \sum_{k=0}^{\infty} \frac{\check{e}^k}{\Gamma(\beta k + 1)}$, where $\check{e}, \beta \in C$ & $R(\beta) > 0$.

If $\beta = 1$, then $E_1(\check{e}) = \sum_{k=0}^{\infty} \frac{\check{e}^k}{\Gamma(k+1)} = \sum_{k=0}^{\infty} \frac{\check{e}^k}{k!} = e^{\check{e}}$.

Throughout this section, we consider $H = \{\digamma : [0,\infty) \to k\}$ denotes the class of all continuously differentiable function of exponential order with $k > 0$ denotes a constant.

Definition 3.2.

Table 1
Shehu Transform of frequently encountered functions

S.No	$Ϝ(\breve{e})$	$S\{Ϝ(\breve{e})\} = Ḥ(v,u)$
1	1	$\frac{u}{v}$
2	$\breve{e}$	$\left(\frac{u}{v}\right)^2$
3	$\breve{e}^2$	$2!\left(\frac{u}{v}\right)^2$
4	$\breve{e}^n, n \in N$	$n!\left(\frac{u}{v}\right)^{n+1}$
5	$\breve{e}^n, n > -1$	$\Gamma(n+1)\left(\frac{u}{v}\right)^{n+1}$
6	$e^{a\breve{e}}$	$\frac{u}{u-au}$
7	$\sin a\breve{e}$	$\frac{au^2}{v^2+a^2u^2}$
8	$\cos a\breve{e}$	$\frac{uv}{(v^2+a^2u^2)}$
9	$\sinh a\breve{e}$	$\frac{au^2}{(v^2-a^2u^2)}$
10	$\cosh l\breve{e}$	$\frac{au}{(v^2-a^2u^2)}$
11	$J_0(a\breve{e})$	$\frac{u}{\sqrt{(v^2+a^2u^2)}}$

Table 2
Inverse Shehu transform of frequently encountered functions

S.No	$Ḥ(u,v)$	$Ϝ(\breve{e}) = s^{-1}\{Ḥ(v,u)\}$
1	$\frac{u}{v}$	1
2	$(\frac{u}{v})^2$	$\breve{e}$
3	$(\frac{u}{v})^3$	$\frac{\breve{e}^2}{2!}$
4	$(\frac{u}{v})^{n+1}, n \in N$	$\frac{\breve{e}^n}{n!}$
5	$(\frac{u}{v})^{n+1}, n > -1$	$\frac{\breve{e}^n}{\Gamma(n+1)}$
6	$\frac{u}{v-au}$	$e^{a\breve{e}}$
7	$\frac{u^2}{(v^2+a^2u^2)}$	$\frac{\sin a\breve{e}}{a}$
8	$\frac{uv}{(v^2+a^2v^2)}$	$\cos a\breve{e}$
9	$\frac{u^2}{(v^2-a^2u^2)}$	$\frac{\sinh a\breve{e}}{a}$
10	$\frac{uv}{(v^2-a^2v^2)}$	$\cosh a\breve{e}$
11	$\frac{u^2}{\sqrt{(v^2-a^2u^2)}}$	$J_0(a\breve{e})$

(i) The differential Eq. (1.3) is said to have the Hyers-Ulam stability in the class H if $\exists\ k > 0$ such that

$$|Ϝ^n(\breve{e}) + a_1(\breve{e})Ϝ^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ϝ'(\breve{e}) + a_n(\breve{e})Ϝ(\breve{e})| \le \epsilon, \quad \epsilon > 0, \ \forall\ \breve{e} \ge 0, \tag{3.1}$$

then there exist a solution $Ç : [0,\infty) \to k$ of (1.3) such that $Ç \in H$ and

$$|Ϝ(\breve{e}) - Ç(\breve{e})| \le k\epsilon.$$

(ii) We say that (1.4) has the Hyers-Ulam stability if $\exists\ k > 0$ such that

$$|Ϝ^n(\breve{e}) + a_1(\breve{e})Ϝ^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ϝ'(\breve{e}) + a_n(\breve{e})Ϝ(\breve{e}) - m(\breve{e})| \le \epsilon, \quad \epsilon > 0, \tag{3.2}$$

for all $\breve{e} \ge 0$, then there exists a solution $Ç : [0,\infty) \to k$ of (1.4) such that $Ç \in H$ and

$$|Ϝ(\breve{e}) - Ç(\breve{e})| \le k\epsilon, \ \forall\ \breve{e} \ge 0,$$

where k is called Hyers-Ulam stability constant (or) stability constant.

Definition 3.3. Let $\sigma : [0,\infty) \to (0,\infty)$ be a function.

(i) We say that (1.3) has the Hyers-Ulam σ-stability (in the class H) if $\exists\ k > 0$ such that

$$|Ϝ^n(\breve{e}) + a_1(\breve{e})Ϝ^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ϝ'(\breve{e}) + a_n(\breve{e})Ϝ(\breve{e})| \le \sigma(\breve{e})\epsilon, \quad \epsilon > 0, \tag{3.3}$$

for all $\check{e} \geq 0$, then there exist a solution $Ç : [0, \infty) \to k$ of (1.3) such that $Ç \in H$ and

$$|Ḟ(\check{e}) - Ç(\check{e})| \leq k\sigma(\check{e})\epsilon, \quad \epsilon > 0, \ \forall \ \check{e} \geq 0.$$

(ii) We say that (1.4) has the Hyers-Ulam σ-stability if $\exists \ k > 0$ such that

$$|Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e}) - m(\check{e})| \leq \sigma(\check{e}), \quad \forall \ \check{e} \geq 0, \tag{3.4}$$

where k is called stability constant.

Definition 3.4. Let $E_\beta(\check{e})$ be the Mittag-Leffler function.

(i) We say that (1.3) has the Mittag-Leffler-Hyers-Ulam stability ($Ḟ \in H$) if their exists a constant $k > 0$ such that

$$|Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e})| \leq E_\beta(\check{e})\epsilon, \quad \epsilon > 0, \tag{3.5}$$

for all $\check{e} \geq 0$, then there exist a solution $Ç : [0, \infty) \to k$ of (1.3) such that $Ç \in H$ and

$$|Ḟ(\check{e}) - Ç(\check{e})| \leq kE_\beta(\check{e})\epsilon, \ \forall \ \check{e} \geq 0.$$

(ii) We say that (1.4) has the Mittag-Leffler-Hyers-Ulam stability ($Ḟ \in H$) if $\exists \ k > 0$ such that

$$|Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e}) - m(\check{e})| \leq E_\beta(\check{e})\epsilon, \quad \epsilon > 0, \tag{3.6}$$

for all $\check{e} \geq 0$, then there exists a solution $Ç : [0, \infty) \to k$ of (1.4) such that $Ç \in H$ and

$$|Ḟ(\check{e}) - Ç(\check{e})| \leq kE_\beta(\check{e})\epsilon, \ \forall \ \check{e} \geq 0,$$

where k is called as Mittag-Leffler-Hyers-Ulam stability constant.

Definition 3.5. Let $E_\beta(\check{e})$ be the Mittag-Leffler function.

(i) We say that (1.3) has the Mittag-Leffler-Hyers-Ulam σ-stability ($Ḟ \in H$), if their exists a constant $k > 0$ such that

$$|Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e})| \leq \sigma(\check{e})E_\beta(\check{e})\epsilon, \quad \epsilon > 0, \tag{3.7}$$

$\forall \ \check{e} \geq 0$, then there exist a solution $Ç : [0, \infty) \to k$ of (1.3) such that $G \in H$ and

$$|Ḟ(\check{e}) - Ç(\check{e})| \leq k\sigma(\check{e})E_\beta(\check{e})\epsilon, \ \forall \ \check{e} \geq 0,$$

(ii) We say that (1.4) has the Mittag-Leffler-Hyers-Ulam σ-stability ($Ḟ \in H$) if $\exists \ k > 0$ such that

$$|Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e}) - m(\check{e})| \leq \sigma(\check{e})E_\beta(\check{e})\epsilon, \quad \epsilon > 0, \tag{3.8}$$

for all $\check{e} \geq 0$, then there exists a solution $Ç : [0, \infty) \to k$ of (1.4) such that $G \in H$ and

$$|Ḟ(\check{e}) - Ç(\check{e})| \leq k\sigma(\check{e})E_\beta(\check{e})\epsilon, \ \forall \ \check{e} \geq 0,$$

where k is called as Mittag-Leffler-Hyers-Ulam σ-stability constant.

Stability of homogeneous linear differential Eq. (1.3)

In this section, we prove several types of Hyers-Ulam stability of the homogeneous linear differential Eq. (1.3) by using the Shehu transform method. Also, we consider $\sigma : [0, \infty) \to (0, \infty)$ is an increasing function.

Theorem 4.1. *Assume that $a_1 + \dots + a_{n-1} + a_n$ is constant with $R(a_1 + \dots + a_{n-1} + a_n) > 0$. Then (1.3) is Hyers-Ulam stable in H.*

Proof. Assume that $Ḟ \in H$ satisfying (3.1), $\forall \ \check{e} \geq 0$ and define the function $i(\check{e}) : [0, \infty) \to k$ by

$$i(\check{e}) = Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e}), \quad \forall \ \check{e} \geq 0. \tag{4.1}$$

The Shehu transform of (4.1) gives the following result:

$$\begin{aligned} S\{i(\check{e})\} = J(v, u) &= S\{Ḟ^n(\check{e}) + a_1(\check{e})Ḟ^{n-1}(\check{e}) + \dots a_{n-1}(\check{e})Ḟ'(\check{e}) + a_n(\check{e})Ḟ(\check{e})\} \\ J(v, u) &= S\{Ḟ^n(\check{e})\} + a_1(\check{e})S\{Ḟ^{n-1}(\check{e})\} + \dots + a_{n-1}(\check{e})S\{Ḟ'(\check{e})\} + a_n(\check{e})S\{Ḟ(\check{e})\}, \end{aligned} \tag{4.2}$$

where $S\{Ḟ(\check{e})\} = Ḥ(v, u)$. Since,

$$S\{Ḟ'(\check{e})\} = \frac{v}{u}Ḥ(v, u) - Ḟ(0)$$

$$S\{Ϝ''(\breve{e})\} = \frac{\nu^2}{u^2}Ḩ(\nu, u) - \frac{\nu}{u}Ϝ(0) - Ϝ'(0)$$

$$\vdots$$

In general for any positive integer n, we have

$$S\{Ϝ^{n-1}(\breve{e})\} = \frac{\nu^{n-1}}{u^{n-1}}Ḩ(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ϝ^{(k)}(0)$$

$$S\{Ϝ^{n}(\breve{e})\} = \frac{\nu^{n}}{u^{n}}Ḩ(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ϝ^{(k)}(0).$$

Now J(v,u) can be written as

$$\begin{aligned} J(\nu, u) = S\{Ϝ^{n}(\breve{e})\} &= \frac{\nu^{n}}{u^{n}}Ḩ(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ϝ^{(k)}(0) \\ &+ a_1(\breve{e})\left(\frac{\nu^{n-1}}{u^{n-1}}Ḩ(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ϝ^{(k)}(0)\right) + \ldots \\ &+ a_{n-1}(\breve{e})\left[\frac{\nu}{u}Ḩ(\nu, u) - Ϝ(0)\right] + a_n(\breve{e})Ḩ_1(\nu, u). \end{aligned}$$

Setting

$$Ḩ_1(\nu, u) = \frac{J(\nu, u) + \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ϝ^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ϝ^{(k)}(0) + \ldots + a_{n-1}(\breve{e})Ϝ(0)}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}. \tag{4.3}$$

If we put $\breve{e} = 0$ in $Ç(\breve{e}) = e^{-(a_1+\ldots+a_{n-1}+a_n)\breve{e}}Ϝ(\breve{e})$, then $Ç(\breve{e}) = Ϝ(\breve{e})$ and $Ç \in H$.

The Shehu transform of $Ç(\breve{e})$ gives the following:

$$\begin{aligned} &Ḩ_2(\nu, u) = S\{Ç(\breve{e})\} \\ &= \frac{\sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ç^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ç^{(k)}(0) + \ldots + a_{n-1}(\breve{e})Ç(0)}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}. \end{aligned} \tag{4.4}$$

Thus,

$$\begin{aligned} &S\{Ç^{n}(\breve{e}) + a_1(\breve{e})Ç^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ç'(\breve{e}) + a_n(\breve{e})Ç(\breve{e})\} \\ &\quad = \frac{\nu^{n}}{u^{n}}Ḩ_2(\nu, u) - \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ç^{(k)}(0) \\ &\quad + a_1(\breve{e})\left(\frac{\nu^{n-1}}{u^{n-1}}Ḩ_2(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ç^{(k)}(0)\right) + \ldots \\ &\quad + a_{n-1}(\breve{e})\left[\frac{\nu}{u}Ḩ_2(\nu, u) - Ç(0)\right] + a_n(\breve{e})Ḩ_2(\nu, u), \quad \forall\ \breve{e} \geq 0. \end{aligned}$$

Then the above equation arrives that

$$\begin{aligned} &\frac{\nu^{n}}{u^{n}}Ḩ_2(\nu, u) - \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}Ç^{(k)}(0) \\ &\quad + a_1(\breve{e})\left(\frac{\nu^{n-1}}{u^{n-1}}Ḩ_2(\nu, u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}Ç^{(k)}(0)\right) + \ldots \\ &\quad + a_{n-1}(\breve{e})\left[\frac{\nu}{u}Ḩ_2(\nu, u) - Ç(0)\right] + a_n(\breve{e})Ḩ_2(\nu, u) = 0 \end{aligned}$$

and

$$\frac{v^n}{u^n}\mathrm{Ḩ}_2(v,u) + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}}\mathrm{Ḩ}_2(v,u) + \ldots + a_{n-1}(\check{e})\frac{v}{u}\mathrm{Ḩ}_2(v,u) + a_n(\check{e})\mathrm{Ḩ}_2(v,u)$$
$$= \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)}\mathrm{Ç}^{(k)}(0) + a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)}\mathrm{Ç}^{(k)}(0) + \ldots + a_{n-1}(\check{e})\mathrm{Ç}(0)$$

$$\mathrm{Ḩ}_2(v,u)\left[\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})\right]$$
$$= \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)}\mathrm{Ç}^{(k)}(0) + a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)}\mathrm{Ç}^{(k)}(0) + \ldots + a_{n-1}(\check{e})\mathrm{Ç}(0).$$

Finally,

$$\mathrm{Ḩ}_2(v,u) = \frac{\sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)}\mathrm{Ç}^{(k)}(0) + a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)}\mathrm{Ç}^{(k)}(0) + \ldots + a_{n-1}(\check{e})\mathrm{Ç}(0)}{\left[\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})\right]}.$$

By (4.4), we have

$$S\{\mathrm{Ç}^n(\check{e}) + a_1(\check{e})\mathrm{Ç}^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})\mathrm{Ç}'(\check{e}) + a_n(\check{e})\mathrm{Ç}(\check{e})\} = 0.$$

Since S is a one - to - one operator. Then

$$\mathrm{Ç}^n(\check{e}) + a_1(\check{e})\mathrm{Ç}^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})\mathrm{Ç}'(\check{e}) + a_n(\check{e})\mathrm{Ç}(\check{e}) = 0.$$

Here $\mathrm{Ç}(\check{e})$ is a solution of (1.3).

By (4.3) and (4.4), we obtain

$$\begin{aligned}S\{\mathrm{F̧}(\check{e})\} - S\{\mathrm{Ç}(\check{e})\} &= \mathrm{Ḩ}_1(v,u) - \mathrm{Ḩ}_2(v,u)\\ &= \frac{J(v,u)}{\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})}\\ &= J(v,u)L(v,u)\\ &= S\{\mathrm{F̧}\star\mathrm{Ç}\}(\check{e})\\ &= [i(\check{e})\star k(\check{e})],\end{aligned}$$

where $L(v,u) = \dfrac{1}{\left(\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})\right)}$ and

$$S\{k(\check{e})\} = L(v,u) = \frac{1}{\left(\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})\right)}$$
$$k(\check{e}) = S^{-1}\left\{\frac{1}{\left(\frac{v^n}{u^n} + a_1(\check{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{v}{u} + a_n(\check{e})\right)}\right\}$$
$$k(\check{e}) = S^{-1}\left\{\frac{u^n}{v^n + a_1(\check{e})v^{n-1} + \ldots + a_{n-1}(\check{e})vu^{n-1} + a_n(\check{e})u^n}\right\}.$$

Then $k(\check{e}) = e^{-(a_1(\check{e})+\ldots+a_{n-1}(\check{e})+a_n(\check{e}))\check{e}}$.

Consequently,

$$S\{\mathrm{F̧}(\check{e}) - \mathrm{Ç}(\check{e})\} = S\{i(\check{e})\star k(\check{e})\}$$

and $\mathrm{F̧}(\check{e}) - \mathrm{Ç}(\check{e}) = i(\check{e})\star k(\check{e})$.

Taking modulus on both sides, we have

$$\begin{aligned}|\mathrm{F̧}(\check{e}) - \mathrm{Ç}(\check{e})| &= |i(\check{e})\star k(\check{e})|\\ &= \left|\int_0^{\check{e}} i(s)k(\check{e}-s)ds\right|\\ &\le \int_0^{\check{e}} |i(s)||k(\check{e}-s)|ds\end{aligned}$$

$$\leq \epsilon \int_0^{\breve{e}} k(\breve{e}-s)ds.$$

Since $k(\breve{e}) = e^{-R(a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e}))\breve{e}}$. Then

$$\begin{aligned}
|Ƒ(\breve{e}) - Ç(\breve{e})| &\leq \epsilon \int_0^{\breve{e}} e^{-R(a_1+\ldots+a_{n-1}+a_n)}(\breve{e}-s)ds \\
&\leq \epsilon \int_0^{\breve{e}} e^{-R(a_1+\ldots+a_{n-1}+a_n)}t.e^{-R(a_1+\ldots+a_{n-1}+a_n)}sds \\
&\leq \epsilon e^{-R(a_1+\ldots+a_{n-1}+a_n)\breve{e}} \int_0^{\breve{e}} e^{-R(a_1+\ldots+a_{n-1}+a_n)}sds \\
&\leq \epsilon e^{-R(a_1+\ldots+a_{n-1}+a_n)\breve{e}} \left[\frac{e^{-R(a_1+\ldots+a_{n-1}+a_n)}\breve{e}-1}{R(a_1+\ldots+a_{n-1}+a_n)}\right] \\
&\leq \epsilon \left[\frac{1-e^{-R(a_1+\ldots+a_{n-1}+a_n)}\breve{e}}{R(a_1+\ldots+a_{n-1}+a_n)}\right] \\
&\leq k\epsilon, \quad \forall\ \breve{e} \geq 0,\ \ k = \frac{1}{R(a_1+\ldots+a_{n-1}+a_n)}.
\end{aligned}$$

This implies that (1.3) has Hyers-Ulam stability in H. □

Note: If $-R(a_1+\ldots+a_{n-1}+a_n) < 0$, then $\frac{\epsilon}{R(a_1+\ldots+a_{n-1}+a_n)}.\left(1-e^{-R(a_1+\ldots+a_{n-1}+a_n)\breve{e}}\right)$ diverges to ∞ as $\breve{e} \to \infty$.
If $-R(a_1+\ldots+a_{n-1}+a_n) < 0$, then we cannot prove Hyers-Ulam stability by using the Shehu transform method.

Corollary 4.1. *Let $\sigma : [0,\infty) \to (0,\infty)$ and $a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e})$ is a constant with $R(a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e})) > 0$. Then (1.3) has the Hyers-Ulam σ-stability in H.*

Proof. Let $Ƒ \in H$, $\sigma : [o,\infty) \to (0,\infty)$ and the inequality (3.3) holds, $\forall\ \breve{e} \geq 0$.
Define a mapping $i : [0,\infty) \to k$ by

$$i(\breve{e}) = Ƒ^n(\breve{e}) + a_1(\breve{e})Ƒ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ƒ'(\breve{e}) + a_n(\breve{e})Ƒ(\breve{e}), \quad \forall\ \breve{e} \geq 0,$$

then we have $|i(\breve{e})| \leq \sigma(\breve{e})\epsilon$, for all $\breve{e} \geq 0$.
The rest of the proof to similar to that of the Theorem 4.1. □

Theorem 4.2. *Let $a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e})$ and β be constants satisfying $R(a_1+\ldots+a_{n-1}+a_n) > 0$ and $\beta > 0$. Then (1.3) has Mittag-Leffler-Hyers-Ulam stability in H.*

Proof. Suppose that $Ƒ \in H$ satisfying (3.5), $\forall\ \breve{e} \geq 0$ and a function $i(\breve{e}) : [0,\infty) \to k$ is defined by

$$i(\breve{e}) = Ƒ^n(\breve{e}) + a_1(\breve{e})Ƒ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ƒ'(\breve{e}) + a_n(\breve{e})Ƒ(\breve{e}), \quad \forall\ \breve{e} \geq 0.$$

By (3.5), we have $|i(\breve{e})| \leq \epsilon$, for all $\breve{e} \geq 0$.
The Shehu transform of $i(\breve{e})$ gives

$$\begin{aligned}
S\{i(\breve{e})\} = J(v,u) &= S\{Ƒ^n(\breve{e}) + a_1(\breve{e})Ƒ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ƒ'(\breve{e}) + a_n(\breve{e})Ƒ(\breve{e})\} \\
J(v,u) &= \frac{v^n}{u^n}Ḩ(v,u) - \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ƒ^{(k)}(0) \\
&+ a_1(\breve{e})\left(\frac{v^{n-1}}{u^{n-1}}Ḩ(v,u) - \sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ƒ^{(k)}(0)\right) + \ldots \\
&+ a_{n-1}(\breve{e})\left[\frac{v}{u}Ḩ(v,u) - Ƒ(0)\right] + a_n(\breve{e})Ḩ_1(v,u).
\end{aligned}$$

Setting

$$Ḩ_1(v,u) = \frac{J(v,u) + \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ƒ^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ƒ^{(k)}(0) + \ldots + a_{n-1}(\breve{e})Ƒ(0)}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}. \tag{4.5}$$

If we put $\c{G}(\breve{e}) = e^{-(a_1+\ldots+a_{n-1}+a_n)\breve{e}}\c{F}(\breve{e})$, then $\c{G}(0) = \c{F}(0)$ and $\c{G} \in H$. The Shehu transform of $\c{G}(\breve{e})$ gives,

$$\begin{aligned}\c{H}_2(\nu,u) &= S\{\c{G}(\breve{e})\}\\ &= \frac{\sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}\c{G}^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}\c{G}^{(k)}(0) + \ldots + a_{n-1}(\breve{e})\c{G}(0)}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}.\end{aligned} \tag{4.6}$$

Thus it follows from (4.6), we obtain

$$\begin{aligned}&S\{\c{G}^n(\breve{e}) + a_1(\breve{e})\c{G}^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})\c{G}'(\breve{e}) + a_n(\breve{e})\c{G}(\breve{e})\}\\ &\quad= \frac{\nu^n}{u^n}\c{H}_2(\nu,u) - \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)}\c{G}^{(k)}(0)\\ &\quad+ a_1(\breve{e})\left(\frac{\nu^{n-1}}{u^{n-1}}\c{H}_2(\nu,u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)}\c{G}^{(k)}(0)\right) + \ldots\\ &\quad+ a_{n-1}(\breve{e})\left[\frac{\nu}{u}\c{H}_2(\nu,u) - \c{G}(0)\right] + a_n(\breve{e})\c{H}_2(\nu,u).\end{aligned}$$

Since S is one to one operator, then

$$\c{G}^n(\breve{e}) + a_1(\breve{e})\c{G}^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})\c{G}'(\breve{e}) + a_n(\breve{e})\c{G}(\breve{e}) = 0.$$

If we set $L(\nu,u) = \frac{1}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}$, which gives

$$\begin{aligned}S\{k(\breve{e})\} &= \frac{1}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}\\ k(\breve{e}) &= S^{-1}\left\{\frac{u^n}{\nu^n + a_1(\breve{e})\nu^{n-1} + \ldots + a_{n-1}(\breve{e})\nu u^{n-1} + a_n(\breve{e})u^n}\right\}\\ k(\breve{e}) &= e^{-(a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e}))\breve{e}}.\end{aligned}$$

Moreover, it follows from (4.5) and (4.6), we obtain

$$\begin{aligned}S\{\c{F}(\breve{e})\} - S\{\c{G}(\breve{e})\} &= \c{H}_1(\nu,u) - \c{H}_2(\nu,u)\\ &= \frac{J(\nu,u)}{\left(\frac{\nu^n}{u^n} + a_1(\breve{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{\nu}{u} + a_n(\breve{e})\right)}\\ &= J(\nu,u)L(\nu,u)\\ &= S\{i(\breve{e}) \star k(\breve{e})\}.\end{aligned} \tag{4.7}$$

This gives

$$\begin{aligned}\c{F}(\breve{e}) - \c{G}(\breve{e}) &= i(\breve{e}) \star k(\breve{e})\\ &= i(\breve{e}) \star e^{-(a_1(\breve{e})+\ldots+a_{n-1}(\breve{e})+a_n(\breve{e}))\breve{e}}.\end{aligned}$$

Taking modulus and using the fact $|i(\breve{e})| \le \epsilon E_\beta(\breve{e})$, for all $\breve{e} \ge 0$.

Since $E_\beta(\breve{e})$ is an increasing function for $\breve{e} \ge 0$ and

$$\begin{aligned}|\c{F}(\breve{e}) - \c{G}(\breve{e})| &= |i(\breve{e}) \star e^{-(a_1+\ldots+a_{n-1}+a_n)\breve{e}}|\\ &= \left|\int_0^{\breve{e}} i(s)e^{-(a_1+\ldots+a_{n-1}+a_n)(\breve{e}-s)}ds\right|\\ &\le \int_0^{\breve{e}} |i(s)||e^{-(a_1+\ldots+a_{n-1}+a_n)(\breve{e}-s)}|ds\\ &\le E_\beta(\breve{e})\epsilon e^{-R(a_1+\ldots+a_{n-1}+a_n)\breve{e}}\int_0^{\breve{e}} e^{R(a_1+\ldots+a_{n-1}+a_n)s}ds\\ &\le \frac{E_\beta(\breve{e})\epsilon}{R(a_1+\ldots+a_{n-1}+a_n)}\left(1 - e^{-R(a_1+\ldots+a_{n-1}+a_n)\breve{e}}\right)\\ &\le kE_\beta(\breve{e})\epsilon, \quad \forall\ \breve{e} \ge 0, \quad k = \frac{1}{R(a_1+\ldots+a_{n-1}+a_n)}.\end{aligned}$$

Then by definition (3.2), we can confirm that (1.3) has Mittag-Leffler-Hyers-Ulam stability in H. □

Corollary 4.2. *Let $\sigma : [0,\infty) \to (0,\infty)$ and $a_1 + \ldots + a_{n-1} + a_n$ and β are constant with $R(a_1 + \ldots + a_{n-1} + a_n) > 0$. Then (1.3) has the Mittag-Leffler-Hyers-Ulam σ-stability in H.*

Proof. Let $Ƒ \in H$, $\sigma : [o,\infty) \to (0,\infty)$ and that $Ƒ(\check{e})$ and $Ç(\check{e})$ satisfy the inequality (3.7), $\forall\ \check{e} \geq 0$.

We will prove that, there exists a constant $k > 0$ (independent of ϵ) such that $Ç : [0,\infty) \to k$ is a solution of (1.3) and $|Ƒ(\check{e}) - Ç(\check{e}) \leq k\sigma(\check{e})\epsilon E_\beta(\check{e})|$, for all $\check{e} \geq 0$.

Define a function $i : [0, infty) \to k$ by

$$i(\check{e}) = Ƒ^n(\check{e}) + a_1(\check{e})Ƒ^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})Ƒ'(\check{e}) + a_n(\check{e})Ƒ(\check{e}), \quad \check{e} \geq 0. \tag{4.8}$$

Then $|i(\check{e})| \leq \sigma(\check{e})\epsilon, \quad \forall\ \check{e} \geq 0$.

The rest of the proof to similar to that of the Theorem 4.2. □

Stability of non-homogeneous linear differential Eq. (1.4)

In this section, we prove several types of Hyers-Ulam stability of the non-homogeneous n^{th} order linear differential Eq.#x00A0; (1.4) by using the Shehu transform. Also, let $m : [0,\infty) \to (0,\infty)$ is a continuous function of exponential order in the class H.

Theorem 5.1. *Let $m : [0,\infty) \to (0,\infty)$ and $a_1 + \ldots + a_{n-1} + a_n$ is a constant with $R(a_1 + \ldots + a_{n-1} + a_n) > 0$. Then (1.3) has the Hyers-Ulam stability in H.*

Proof. Suppose that $Ƒ \in H$ satisfying (3.2), $\forall\ \check{e} \geq 0$.

Define the function $i : [0,\infty) \to k$ by

$$i(\check{e}) = Ƒ^n(\check{e}) + a_1(\check{e})Ƒ^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})Ƒ'(\check{e}) + a_n(\check{e})Ƒ(\check{e}) - m(\check{e}), \quad \forall\ \check{e} \geq 0. \tag{5.1}$$

Then $|i(\check{e})| \leq \epsilon$ holds, $\forall\ \check{e} \geq 0$.

The Shehu transform of the function $i(\check{e})$ gives

$$S\{i(\check{e})\} = J(\nu,u) = S\{Ƒ^n(\check{e}) + a_1(\check{e})Ƒ^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})Ƒ'(\check{e}) + a_n(\check{e})Ƒ(\check{e}) - m(\check{e})\}. \tag{5.2}$$

This implies that,

$$S\{Ƒ(\check{e})\} = Ḩ_1(\nu,u) = \frac{J(\nu,u) + \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)} Ƒ^{(k)}(0) + a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)} Ƒ^{(k)}(0) + \ldots + a_{n-1}(\check{e})Ƒ(0)}{\left(\frac{\nu^n}{u^n} + a_1(\check{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{\nu}{u} + a_n(\check{e})\right)}. \tag{5.3}$$

If we set $Ç(\check{e}) = e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}Ƒ(\check{e}) + (m(\check{e}) \star e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}})$, then $Ç(0) = Ƒ(0)$ and $Ç \in H$.

The Shehu transform of $Ç(\check{e})$ gives the following result:

$$S\{Ç(\check{e})\} = Ḩ_2(\nu,u) = \frac{\sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)} Ç^{(k)}(0) + a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)} Ç^{(k)}(0) + \ldots + a_{n-1}(\check{e})Ç(0)}{\left(\frac{\nu^n}{u^n} + a_1(\check{e})\frac{\nu^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\check{e})\frac{\nu}{u} + a_n(\check{e})\right)}. \tag{5.4}$$

On the other hand

$$\begin{aligned}
&S\{Ç^n(\check{e}) + a_1(\check{e})Ç^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})Ç'(\check{e}) + a_n(\check{e})Ç(\check{e})\} \\
&= \frac{\nu^n}{u^n}Ḩ_2(\nu,u) - \sum_{k=0}^{n-1}\left(\frac{\nu}{u}\right)^{n-(k+1)} Ç^{(k)}(0) \\
&+ a_1(\check{e})\left(\frac{\nu^{n-1}}{u^{n-1}}Ḩ_2(\nu,u) - \sum_{k=0}^{n-2}\left(\frac{\nu}{u}\right)^{(n-1)-(k+1)} Ç^{(k)}(0)\right) + \ldots \\
&+ a_{n-1}(\check{e})\left[\frac{\nu}{u}Ḩ_2(\nu,u) - Ç(0)\right] + a_n(\check{e})Ḩ_2(\nu,u) >
\end{aligned}$$

Using (5.4), we have

$$S\{Ç^n(\check{e}) + a_1(\check{e})Ç^{n-1}(\check{e}) + \ldots a_{n-1}(\check{e})Ç'(\check{e}) + a_n(\check{e})Ç(\check{e})\} = S\{m(\check{e})\} = N(\nu,u)$$

and

$$Ç^{n}(\check{e})+a_1(\check{e})Ç^{n-1}(\check{e})+\ldots a_{n-1}(\check{e})Ç'(\check{e})+a_n(\check{e})Ç(\check{e})=m(\check{e}).$$

Hence, $Ç(\check{e})$ is a solution of (1.3).

By applying (5.3) and (5.4), we can obtain

$$\begin{aligned}S\{Ƒ(\check{e})\}-S\{Ç(\check{e})\}&=Ḩ_1(v,u)-Ḩ_2(v,u)\\&=\frac{J(v,u)}{\frac{v^n}{u^n}+a_1(\check{e})\frac{v^{n-1}}{u^{n-1}}+\ldots+a_{n-1}(\check{e})\frac{v}{u}+a_n(\check{e})}\\&=J(v,u)L(v,u)\\&=S\{i(\check{e})\star k(\check{e})\},\end{aligned}$$

where $L(v,u)=S\{k(\check{e})\}=\frac{1}{\left(\frac{v^n}{u^n}+a_1(\check{e})\frac{v^{n-1}}{u^{n-1}}+\ldots+a_{n-1}(\check{e})\frac{v}{u}+a_n(\check{e})\right)}$ and

$$\begin{aligned}k(\check{e})&=S^{-1}\left\{\frac{u^n}{v^n+a_1(\check{e})v^{n-1}+\ldots+a_{n-1}(\check{e})vu^{n-1}+a_n(\check{e})u^n}\right\}.\\k(\check{e})&=e^{-(a_1(\check{e})+\ldots+a_{n-1}(\check{e})+a_n(\check{e}))\check{e}}.\end{aligned}$$

Therefore,

$$\begin{aligned}S\{Ƒ(\check{e})-Ç(\check{e})\}&=S\{i(\check{e})\star k(\check{e})\}\\&=S\{i(\check{e})\star e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}\}.\end{aligned}$$

Which yields $Ƒ(\check{e})-Ç(\check{e})=i(\check{e})\star e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}$.

Furthermore,

$$\begin{aligned}|Ƒ(\check{e})-Ç(\check{e})|&=|i(\check{e})\star|e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}|\\&=|i(s)e^{-(a_1+\ldots+a_{n-1}+a_n)(\check{e}-s)ds}|\\&\le\int_0^{\check{e}}|i(s)||e^{-(a_1+\ldots+a_{n-1}+a_n)(\check{e}-s)}|ds\\&\le\epsilon e^{-R(a_1+\ldots+a_{n-1}+a_n)\check{e}}\int_0^{\check{e}}e^{R(a_1+\ldots+a_{n-1}+a_n)s}ds\\&\le\frac{\epsilon}{R(a_1+\ldots+a_{n-1}+a_n)}\left(1-e^{-R(a_1+\ldots+a_{n-1}+a_n)\check{e}}\right)\\&\le k\epsilon,\quad\forall\ \check{e}\ge0,\end{aligned}$$

where we set $k=\frac{1}{R(a_1+\ldots+a_{n-1}+a_n)}$. □

Theorem 5.2. *Let $m:[0,\infty)\to k$ is a continuous function, $\sigma:[0,\infty)\to(0,\infty)$ is an increasing function and $a_1+\ldots+a_{n-1}+a_n$ is a constant with $R(a_1+\ldots+a_{n-1}+a_n)>0$. Then (1.4) has the Hyers-Ulam σ-stability in H.*

Proof. Suppose that the function $Ƒ\in H$ satisfying (3.4), $\forall\ \check{e}\ge0$.

If we define a function $i:[0,\infty)\to k$ by

$$i(\check{e})=Ƒ^{n}(\check{e})+a_1(\check{e})Ƒ^{n-1}(\check{e})+\ldots a_{n-1}(\check{e})Ƒ'(\check{e})+a_n(\check{e}),Ƒ(\check{e})-m(\check{e}),\quad\forall\ \check{e}\ge0,$$

then $|i(\check{e})|\le\sigma(\check{e})\epsilon$, for all $\check{e}\ge0$.

It is easy to see that

$$S\{Ƒ(\check{e})\}=Ḩ_1(v,u)=\frac{J(v,u)+\sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)}Ƒ^{(k)}(0)+a_1(\check{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)}Ƒ^{(k)}(0)+\ldots+a_{n-1}(\check{e})Ƒ(0)}{\left(\frac{v^n}{u^n}+a_1(\check{e})\frac{v^{n-1}}{u^{n-1}}+\ldots+a_{n-1}(\check{e})\frac{v}{u}+a_n(\check{e})\right)}.\tag{5.5}$$

If we set $Ç(\check{e})=e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}Ƒ(\check{e})+(m(\check{e})\star e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}})$, then $Ç\in H$.

Further, we apply the Shehu transform to get,

$$S\{Ç(\check{e})\}=Ḩ_2(v,u)$$

$$= \frac{\sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ģ^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ģ^{(k)}(0) + \ldots + a_{n-1}(\breve{e})Ģ(0)}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}. \tag{5.6}$$

On the other hand

$$\begin{aligned} &S\{Ģ^n(\breve{e}) + a_1(\breve{e})Ģ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ģ'(\breve{e}) + a_n(\breve{e})Ģ(\breve{e})\} \\ &\quad = \frac{v^n}{u^n}Ḩ_2(v,u) - \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ģ^{(k)}(0) \\ &\quad + a_1(\breve{e})\left(\frac{v^{n-1}}{u^{n-1}}Ḩ_2(v,u) - \sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ģ^{(k)}(0)\right) + \ldots \\ &\quad + a_{n-1}(\breve{e})\left[\frac{v}{u}Ḩ_2(v,u) - Ģ(0)\right] + a_n(\breve{e})Ḩ_2(v,u). \end{aligned}$$

From the relation (5.6), we have

$$S\{Ģ^n(\breve{e}) + a_1(\breve{e})Ģ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ģ'(\breve{e}) + a_n(\breve{e})Ģ(\breve{e})\} = S\{m(\breve{e})\} = N(v,u)$$

and

$$Ģ^n(\breve{e}) + a_1(\breve{e})Ģ^{n-1}(\breve{e}) + \ldots a_{n-1}(\breve{e})Ģ'(\breve{e}) + a_n(\breve{e})Ģ(\breve{e}) = m(\breve{e}).$$

That is $Ģ(\breve{e})$ is a solution of (1.4). Using (5.5) and (5.6), we obtain

$$\begin{aligned} S\{F̧(\breve{e})\} - S\{Ģ(\breve{e})\} &= Ḩ_1(v,u) - Ḩ_2(v,u) \\ &= \frac{J(v,u)}{\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})} \\ &= J(v,u)L(v,u) \\ &= S\{i(\breve{e}) \star k(\breve{e})\}, \end{aligned}$$

where $L(v,u) = S\{k(\breve{e})\} = \frac{1}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \ldots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}$.

This result gives

$$k(\breve{e}) = S^{-1}\left\{\frac{u^n}{v^n + a_1(\breve{e})v^{n-1} + \ldots + a_{n-1}(\breve{e})vu^{n-1} + a_n(\breve{e})u^n}\right\}.$$

$$k(\breve{e}) = e^{-(a_1(\breve{e}) + \ldots + a_{n-1}(\breve{e}) + a_n(\breve{e}))\breve{e}}.$$

Therefore,

$$S\{F̧(\breve{e}) - Ģ(\breve{e})\} = S\{i(\breve{e}) \star k(\breve{e})\},$$

which gives $F̧(\breve{e}) - Ģ(\breve{e}) = i(\breve{e}) \star k(\breve{e})$.

By Theorem (4.3), we have

$$\begin{aligned} |F̧(\breve{e}) - Ģ(\breve{e})| &= |i(\breve{e}) \star k(\breve{e})| \\ &= |i(s) \star k(\breve{e} - s)ds| \\ &\leq \int_0^{\breve{e}} |i(s)||e^{-(a_1 + \ldots + a_{n-1} + a_n)(\breve{e}-s)ds}| \\ &\leq \sigma(\breve{e})\epsilon \int_0^{\breve{e}} e^{-(a_1 + \ldots + a_{n-1} + a_n)(\breve{e}-s)}|ds \\ &\leq \sigma(\breve{e})\epsilon e^{-R(a_1 + \ldots + a_{n-1} + a_n)\breve{e}} \int_0^{\breve{e}} e^{R(a_1 + \ldots + a_{n-1} + a_n)s}ds \\ &\leq k\sigma(\breve{e})\epsilon, \quad \forall\ \breve{e} \geq 0, \end{aligned}$$

where we set $k = \frac{1}{R(a_1 + \ldots + a_{n-1} + a_n)}$. □

Theorem 5.3. *Let $m : [0, \infty) \to (0, \infty)$ is an continuous function and $a_1 + \ldots + a_{n-1} + a_n$, $\beta > 0$ are constants satisfying $R(a_1 + \ldots + a_{n-1} + a_n) > 0$. Then (1.4) has Mittag-Leffler-Hyers-Ulam stability in H.*

Proof. Suppose that $Ƒ \in H$ and $Ƒ(\breve{e})$ satisfying (3.6), $\forall\ \breve{e} \geq 0$.

A function $i : [0, \infty) \to k$ is defined by

$$i(\breve{e}) = Ƒ^n(\breve{e}) + a_1(\breve{e})Ƒ^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ƒ'(\breve{e}) + a_n(\breve{e})Ƒ(\breve{e}) - m(\breve{e}), \quad \forall\ \breve{e} \geq 0.$$

It follows from (3.6), we obtain $|i(\breve{e})| \leq E_\beta(\breve{e})\epsilon,\ \forall\ \breve{e} \geq 0$.

The Shehu transform of a function $i(\breve{e})$ gives

$$S\{i(\breve{e})\} = J(v,u) = S\{Ƒ^n(\breve{e}) + a_1(\breve{e})Ƒ^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ƒ'(\breve{e}) + a_n(\breve{e})Ƒ(\breve{e}) - m(\breve{e})\}.$$

That is,

$$S\{Ƒ(\breve{e})\} = Ḩ_1(v,u) = \frac{J(v,u) + \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ƒ^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ƒ^{(k)}(0) + \dots + a_{n-1}(\breve{e})Ƒ(0)}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \dots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}. \tag{5.7}$$

If we set $Ç(\breve{e}) = e^{-(a_1+\dots+a_{n-1}+a_n)\breve{e}}Ƒ(\breve{e}) + (m(\breve{e}) \star e^{-(a_1+\dots+a_{n-1}+a_n)\breve{e}})$, then $Ç(0) = Ƒ(0)$ and $Ƒ \in H$.

We apply the Shehu transform of last equality to get

$$S\{Ç(\breve{e})\} = Ḩ_2(v,u) = \frac{\sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ç^{(k)}(0) + a_1(\breve{e})\sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ç^{(k)}(0) + \dots + a_{n-1}(\breve{e})Ç(0)}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \dots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}. \tag{5.8}$$

On the other hand

$$\begin{aligned} &S\{Ç^n(\breve{e}) + a_1(\breve{e})Ç^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ç'(\breve{e}) + a_n(\breve{e})Ç(\breve{e})\} \\ &\quad = \frac{v^n}{u^n}Ḩ_2(v,u) - \sum_{k=0}^{n-1}\left(\frac{v}{u}\right)^{n-(k+1)} Ç^{(k)}(0) \\ &\quad + a_1(\breve{e})\left(\frac{v^{n-1}}{u^{n-1}}Ḩ_2(v,u) - \sum_{k=0}^{n-2}\left(\frac{v}{u}\right)^{(n-1)-(k+1)} Ç^{(k)}(0)\right) + \dots \\ &\quad + a_{n-1}(\breve{e})\left[\frac{v}{u}Ḩ_2(v,u) - Ç(0)\right] + a_n(\breve{e})Ḩ_2(v,u). \end{aligned}$$

By (5.8), we have

$$S\{Ç^n(\breve{e}) + a_1(\breve{e})Ç^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ç'(\breve{e}) + a_n(\breve{e})Ç(\breve{e})\} = S\{m(\breve{e})\} = N(v,u)$$

and

$$Ç^n(\breve{e}) + a_1(\breve{e})Ç^{n-1}(\breve{e}) + \dots a_{n-1}(\breve{e})Ç'(\breve{e}) + a_n(\breve{e})Ç(\breve{e}) = m(\breve{e}).$$

Hence, $Ç(\breve{e})$ is a solution of (1.4).

By applying (5.7) and (5.8), we can obtain

$$\begin{aligned} S\{Ƒ(\breve{e})\} - S\{Ç(\breve{e})\} &= Ḩ_1(v,u) - Ḩ_2(v,u) \\ &= \frac{J(v,u)}{\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \dots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})} \\ &= J(v,u)L(v,u) \\ &= S\{i(\breve{e}) \star k(\breve{e})\}, \end{aligned} \tag{5.9}$$

where $L(v,u) = S\{k(\breve{e})\} = \frac{1}{\left(\frac{v^n}{u^n} + a_1(\breve{e})\frac{v^{n-1}}{u^{n-1}} + \dots + a_{n-1}(\breve{e})\frac{v}{u} + a_n(\breve{e})\right)}$.

This gives,

$$k(\breve{e}) = S^{-1}\left\{\frac{u^n}{v^n + a_1(\breve{e})v^{n-1} + \dots + a_{n-1}(\breve{e})vu^{n-1} + a_n(\breve{e})u^n}\right\}.$$

$$k(\breve{e}) = e^{-(a_1(\breve{e})+\dots+a_{n-1}(\breve{e})+a_n(\breve{e}))\breve{e}}. \tag{5.10}$$

Therefore,

$$S\{Ƒ(\check{e}) - Ç(\check{e})\} = S\{i(\check{e}) \star k(\check{e})\},$$

and $Ƒ(\check{e}) - Ç(\check{e}) = i(\check{e}) \star k(\check{e})$ for all $\check{e} \geq 0$.

Furthermore,

$$\begin{aligned}|Ƒ(\check{e}) - Ç(\check{e})| &= |i(\check{e}) \star |e^{-(a_1+\ldots+a_{n-1}+a_n)\check{e}}| \\ &= |i(s)e^{-(a_1+\ldots+a_{n-1}+a_n)(\check{e}-s)ds}| \\ &\leq \int_0^{\check{e}} |i(s)||e^{-(a_1+\ldots+a_{n-1}+a_n)(\check{e}-s)|ds} \\ &\leq \frac{E_\beta(\check{e})\epsilon}{R(a_1+\ldots+a_{n-1}+a_n)}\left(1 - e^{-R(a_1+\ldots+a_{n-1}+a_n)\check{e}}\right) \\ &\leq kE_\beta(\check{e})\epsilon, \quad \forall\ \check{e} \geq 0,\end{aligned}$$

where $k = \frac{1}{R(a_1+\ldots+a_{n-1}+a_n)}$. This completes the proof. □

Corollary 5.1. *Let $m : [0, \infty) \to (0, \infty)$ is an continuous function, $\sigma; [0, \infty) \to (0, \infty)$ is an increasing function and $a_1 + \ldots + a_{n-1} + a_n$, $\beta > 0$ are constants which satisfy $R(a_1 + \ldots + a_{n-1} + a_n) > 0$. Then (1.4) has the Mittag-Leffler-Hyers-Ulam σ-stability in H.*

Proof. The rest of the proof to similar to that of the Theorem 5.3. □

Application of Shehu transform

In this section, we examine some applications of the Shehu transform method.

Shehu transform of fractional differential equations

We consider the following general linear differential equation of fractional order:

$${}^cD^{\mu}_{0^+}(\check{e}) = \sum_{i=0}^{n} \check{b}_i y^{(i)}(\check{e}) + g(\check{e}), \quad n-1 < \mu \leq n \tag{6.1}$$

subject to the initial condition

$$y^{(i)}(0) = \check{a}_i, \quad i = 0, ..., n-1, \tag{6.2}$$

where $\check{a}_i, \check{b}_j \in R$.

We obtain Shehu transform of (6.1) as follows

$$Ş({}^cD^{\mu}_{0^+}(\check{e})) = Ş\left(\sum_{i=0}^{n} \check{b}_i y^{(i)}(\check{e}) + g(\check{e})\right), \quad n-1 < \mu \leq n.$$

By the linearity of Shehu transform, we have

$$\begin{aligned}Ş({}^cD^{\mu}_{0^+}(\check{e})) &= Ş\left(\sum_{i=0}^{n} \check{b}_i y^{(i)}(\check{e})\right) + Ş\big(g(\check{e})\big), \quad n-1 < \mu \leq n \\ &= \check{b}_0 y(\check{e}) + \sum_{i=1}^{n} \check{b}_i Ş\big(y^{(i)}(\check{e})\big) + Ş\big(g(\check{e})\big).\end{aligned}$$

Using Shehu transform of derivatives, we obtain

$$\left(\frac{u}{v}\right)^{\mu} Ḩ(v,u) - \sum_{k=0}^{n-1}\left(\frac{u}{v}\right)^{k+1-\mu} y^{(k)}(0) = \check{b}_0 Ḩ(v,u) + \sum_{i=1}^{n} \check{b}_i \left[\left(\frac{u}{v}\right)^{-i} Ḩ(v,u) - \sum_{k=0}^{i-1}\left(\frac{u}{v}\right)^{k+1-i} y^{(k)}(0)\right] + Ş\big(g(\check{e})\big)$$

$$\left(\frac{u}{v}\right)^{\mu} Ḩ(v,u) - \sum_{i=0}^{n-1} \check{b}_i \left(\frac{u}{v}\right)^{-i} Ḩ(v,u) = \sum_{k=0}^{n-1} \check{a}_k \left(\frac{u}{v}\right)^{k+1-\mu} - \sum_{i=1}^{n} \check{b}_i \sum_{k=0}^{i-1} \check{a}_k \left(\frac{u}{v}\right)^{k+1-i} + Ş\big(g(\check{e})\big)$$

$$Ḩ(v,u) = \left(\left(\frac{u}{v}\right)^{\mu} - \sum_{i=0}^{n-1} \check{b}_i \left(\frac{u}{v}\right)^{-i}\right)^{-1} \left(\sum_{k=0}^{n-1} \check{a}_k \left(\frac{u}{v}\right)^{k+1-\mu} - \sum_{i=1}^{n} \check{b}_i \sum_{k=0}^{i-1} \check{a}_k \left(\frac{u}{v}\right)^{k+1-i} + Ş\big(g(\check{e})\big)\right). \tag{6.3}$$

Operating the inverse Shehu transform on both sides of Eq. (6.3), we get the solution of Eq. (6.1) as follows:

$$y(\check{e}) = \c{S}^{-1}\left[\left(\left(\frac{u}{v}\right)^{\mu} - \sum_{i=0}^{n-1}\check{b}_i\left(\frac{u}{v}\right)^{-i}\right)^{-1}\left(\sum_{k=0}^{n-1}\breve{a}_k\left(\frac{u}{v}\right)^{k+1-\mu} - \sum_{i=1}^{n}\check{b}_i\sum_{k=0}^{i-1}\breve{a}_k\left(\frac{u}{v}\right)^{k+1-i} + \c{S}\big(g(\check{e})\big)\right)\right]. \tag{6.4}$$

Remark 6.1. When $n = 1, \check{b}_0 = -1$ and $\check{b}_1 = g(\check{e}) = 0$, we obtain

$${}^{c}D^{\mu}y(\check{e}) + y(\check{e}) = 0, \quad 0 < \mu \le 1, \ \check{e} > 0 \tag{6.5}$$

with initial condition

$$y(0) = 1. \tag{6.6}$$

Substituting $n, \check{b}_0, \check{b}_1$ and g in (6.4), we get

$$\begin{aligned} y(\check{e}) &= \c{S}^{-1}\left[\left(\left(\frac{u}{v}\right)^{-\mu} - \sum_{i=0}^{n-1}\check{b}_i\left(\frac{u}{v}\right)^{-i}\right)^{-1}\left(\frac{u}{v}\right)^{1-\mu}\right] \\ &= \c{S}^{-1}\left[\left(\frac{u}{v}\right) - \left(1 - (-1)\left(\frac{u}{v}\right)\right)^{-1}\right]. \end{aligned}$$

Thus, by definition 3.1, we have

$$\c{H}(v,u) = \c{S}(E_{\mu}(-\check{e}^{\mu})). \tag{6.7}$$

When we get the inverse Shehu transform of (6.7), we find exact solution of Eq. (6.5) as follows:

$$y(\check{e}) = E_{\mu}(-\check{e}^{\mu}).$$

Remark 6.2. Below we give the following particular example, which where debate in the literature and here important application in several world problems.

Consider the Bagley-Torvik equation

$$D^2y(\check{e}) +^{c} D^{\frac{3}{2}}y(\check{e}) + y(\check{e}) = \check{e} + 1, \tag{6.8}$$

with the initial conditions

$$y(0) = y'(0) = 1. \tag{6.9}$$

In this case, we have $n = 2, \check{b}_0 = \check{b}_2 = -1, \check{b}_1 = 0$ and $g(\check{e}) = \check{e} + 1$. Applying Eq. (6.4), we get

$$y(\check{e}) = \c{S}^{-1}\left[\left(\left(\frac{u}{v}\right)^{-\frac{3}{2}} - \sum_{i=0}^{2}\check{b}_i\left(\frac{u}{v}\right)^{-i}\right)^{-1}\left(\sum_{k=0}^{1}\breve{a}_k\left(\frac{u}{v}\right)^{k+1-\frac{3}{2}} - \sum_{i=1}^{2}\check{b}_i\sum_{k=0}^{i-1}\breve{a}_k\left(\frac{u}{v}\right)^{k+1-i} + \left(\frac{u}{v}\right) + \left(\frac{u}{v}\right)^{2}\right)\right]. \tag{6.10}$$

Then

$$\begin{aligned} y(\check{e}) &= \c{S}^{-1}\left(\frac{\left(\frac{u}{v}\right)^{-\frac{1}{2}} + \left(\frac{u}{v}\right)^{\frac{1}{2}} + \left(\frac{u}{v}\right)^{-1} + 1 + \left(\frac{u}{v}\right) + \left(\frac{u}{v}\right)^{2}}{\left(\frac{u}{v}\right)^{-\frac{3}{2}} + \left(\frac{u}{v}\right)^{-2} + 1}\right) \\ &= \c{S}^{-1}\left(\frac{\left(\frac{u}{v}\right)^{-\frac{1}{2}} + \left(\frac{u}{v}\right)^{\frac{1}{2}} + \left(\frac{u}{v}\right)^{-1}}{\left(\frac{u}{v}\right)^{-\frac{3}{2}} + \left(\frac{u}{v}\right)^{-2} + 1} + \frac{1 + \left(\frac{u}{v}\right) + \left(\frac{u}{v}\right)^{2}}{\left(\frac{u}{v}\right)^{-\frac{3}{2}} + \left(\frac{u}{v}\right)^{-2} + 1}\right) \\ &= \c{S}^{-1}\left(\left(\frac{u}{v}\right) + \left(\frac{u}{v}\right)^{2}\right). \end{aligned} \tag{6.11}$$

Taking the inverse Shehu transform of Eq. (6.11), yields

$$y(\check{e}) = \check{e} + 1,$$

which is the exact solution.

Shehu Transform for handling Newton's law of cooling problem

The temperature of an object of mass M heated by a flame and cooled by the surrounding medium is computed using Newton's Law of Cooling. The model assumes that the object's temperature, T, is constant. If the rate of thermal energy transfer within the item is faster than the rate of thermal energy transfer at the surface, then this lumped system approximation is true. According to Newton's law of cooling, the pace at which a warm body cools is approximately equal to the difference between the warm object's temperature and the temperature of its environment.

Newton's rule of cooling is usually limited to simple instances in which energy is transferred by convection from a solid surface to a moving fluid, and the temperature difference is modest (less than $10^{\circ}C$) (The Encyclopedia Britannica, 1910). When the medium in which the heated body is placed changes from a simple fluid to a gas, solid, or vacuum, for example, this becomes a residual effect that requires further investigation (Whewell, 1866).

Newtons law of cooling based on the assumptions above stated mathematically as

$$\frac{dT}{dt} = -k(T - T_s),$$

where

T: Temperature of the cooling object at time t t: time in hours

Ts: Temperature of surrounding medium (Ambient temperature) k: Constant of proportionality

Solution of Newton's Equation

$$\frac{dT}{dt} = -k(T - Ts) \tag{6.12}$$

$$\frac{dT}{dt} + kT = kTs. \tag{6.13}$$

Taking Shehu transform both sides of Eq. (6.13), we get

$$S\left[\frac{dT}{dt} + kT\right] = S[kTs]$$

$$S\left[\frac{dT}{dt}\right] + kS[T] = kTsS[1]$$

$$\frac{s}{u}Ƒ(s,u) + kƑ(s,u) = T(0) + kTs\frac{s}{u}$$

$$Ƒ(s,u) = \frac{T(0)}{\frac{s}{u} + k} + \frac{kTs\frac{s}{u}}{\frac{s}{u} + k}$$

$$Ƒ(s,u) = T(0)\frac{u}{s+uk} + Ts\frac{u}{s} - Ts\frac{u}{s+uk}.$$

Applying the inverse transform gives the solution:

$$T(t) = T(0) + e^{-kt} - Tse^{-kt} = T(0) + (1 - Ts)e^{-kt}. \tag{6.14}$$

Remark 6.3. The rate of cooling of a body is proportional to the difference between the temperature of the body and the surrounding air. If the air temperature is $20^{\circ}C$ and the body cools for 20 minutes from $140^{\circ}C$ to $80^{\circ}C$, find when the temperature will be $35^{\circ}C$.

Then from Newton's law of cooling, (6.14) becomes

$$\frac{dT}{dt} + kT = 20k.$$

Take the Shehu Transform both sides, we obtain

$$T(t) = 20 + 120e^{-kt}.$$

Now, using the condition at time t=20, i.e $T(28) = 80$. It follows that

$$k = 0.034657.$$

Now substituting the value of k, we get

$$T(t) = 20 + 120e^{-0.034657t}. \tag{6.15}$$

In Fig. 1, represents the graphical solution of the problem (6.14).

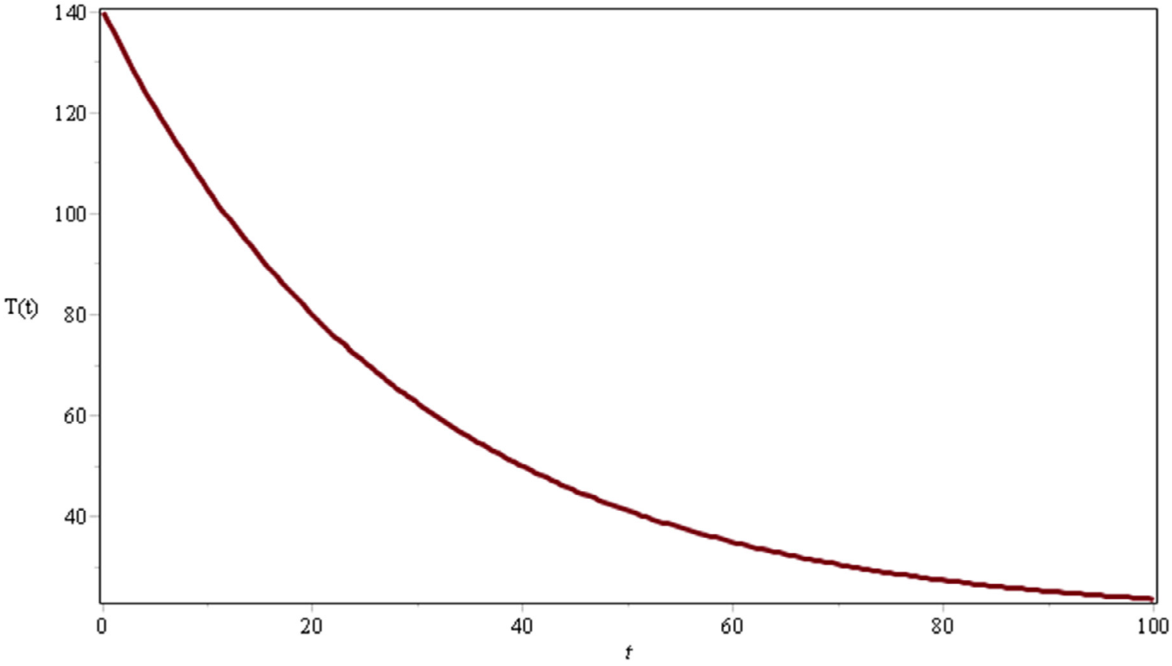

Fig. 1. Solution curve (6.15)

Shehu transform for handling free undamped motion problem

Hooke's Law states that if a spring is stretched (or compressed) x units from its natural length, it exerts a force proportional to x. If we ignore any external resistive forces (such as air resistance or friction), we may apply Newton's Second Law (force equals mass times acceleration/n) to arrive at the following equation:

$$m\frac{d^2x}{dt^2} = -k(x+s) + mg = -ks + mg - ks, \tag{6.16}$$

where k is a positive constant (called the spring constant).

Since, $mg - ks = 0$ ($\because$ Newton second law). Therefore Eq. (6.16) is equal to:

$$m\frac{d^2x}{dt^2} = -ks. \tag{6.17}$$

Free Undamped Motion: By dividing (6.17) by the mass m, we obtain the second order differential equation $m\frac{d^2x}{dt^2} = -\frac{k}{m}s$, this can be written as:

$$m\frac{d^2x}{dt^2} = -\omega s, \tag{6.18}$$

where $\frac{k}{m}$.

Eq. (6.18) is said to describe simple harmonic motion or free undamped motion.

Taking Shehu transform both sides of (6.18), we get

$$Ꞙ(s,u) = \frac{v(0)su}{s^2+\omega^2u^2} + \frac{v'(0)}{\omega}\frac{\omega u^2}{s^2+\omega^2u^2}.$$

Applying the inverse transform gives the solution

$$Ꞙ(s,u) = c_1\cos\omega t + c_2\sin\omega t,$$

where $c_1 = v(0),\ c_2 = \frac{v'(0)}{\omega}$.

Remark 6.4. Consider a spring with a mass of 2 kg has natural length 0.5 m. a force of 25.6 N is required to maintain it stretched to length of 0.7 m. If the spring is stretched to a length of 0.7m and then released with initial velocity 0, find the position of the mass at any time t.

From hook's law the force required to stretched the spring is $k(0.2) = 25.6$. So, $k = \frac{25.6}{0.2} = 128$.

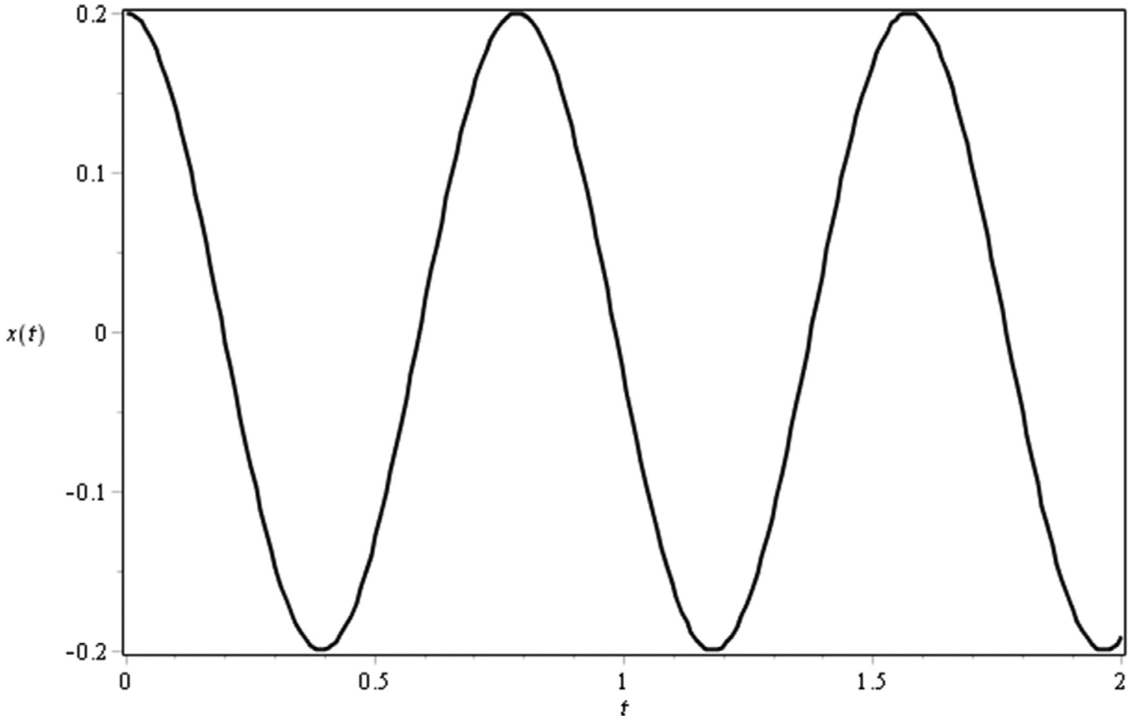

Fig. 2. Solution curve $x(t) = 0.2\cos(8t)$.

Using this value of the spring constant k, together with $m = 2$ in the problem (6.16), we have

$$\frac{d^2x}{dt^2} + 64x = 0.$$

Now, taking the Shehu Transform both sides

$$Ḟ(s,u) = \frac{x(0)su}{s^2 + (8u)^2} + \frac{1}{8}x'(0)\frac{8u^2}{s^2 + (8u)^2}.$$

Taking the inverse of Shehu Transform, we get

$$x(t) = x(0)\cos(8t) + \frac{1}{8}x'(0)\sin(8t).$$

We are given initial condition that $x(0) = 0.2$ and the initial velocity given as $x'(0) = 0$, therefore our solution is converted to:

$$x(t) = 0.2\cos(8t).$$

Shehu transform for handling free undamped motion problem is graphically represented in Fig. 2.

Conclusion

In the field of differential calculus, finding a new integral transform for solving the higher order Shehu transform of a linear differential equation is always useful. In this manuscript, the newly suggested Shehu integral transform was applied to solve the Hyers-Ulam stability of the linear differential equation with Shehu sense. We show the efficiency and high accuracy of the suggested integral transform. Finally, the applications and comments are discussed to demonstrate our strategy. Finally, applications of the Shehu transform to fractional differential equations, Newton's law of cooling, and free undamped motion are also discussed.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

ORNL-DAAC Reference Title

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